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DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

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ANN - Lab

Project # RoomMind AI

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1. INTRODUCTION

People may not immediately notice when something in their room has been moved, removed, or added. Comparing photographs manually can be time-consuming, especially when a room contains many objects.

RoomMind AI is a web application designed to compare two photographs of the same room: one taken before leaving and another taken after returning. It aims to identify possible changes, classify them as added, removed, moved, or unchanged, and highlight the affected regions.

The project uses Python, OpenCV, and a Siamese Convolutional Neural Network (CNN) trained from scratch. A Flask backend will connect the trained model to a website developed using HTML, CSS, and JavaScript.

1.1 Problem Statement

Comparing two room photographs is challenging because lighting changes, shadows, and camera movement can make an unchanged room appear different. Basic image-comparison methods may therefore produce false detections.

RoomMind AI aims to develop a trained CNN that identifies meaningful changes between two photographs, classifies them into predefined categories, and highlights their locations. Image preprocessing and alignment will be used to reduce unnecessary differences.

1.2 Objectives

1. Prepare a labelled dataset of room photographs.
2. Apply image preprocessing and alignment using OpenCV.
3. Train a Siamese CNN from scratch using labelled image pairs.
4. Identify and classify added, removed, moved, and unchanged scenes or regions.
5. Develop a web application for uploading photographs and displaying results.
6. Evaluate the model using suitable performance metrics.
7. Compare the trained model with a basic OpenCV image-difference method.

2. PROPOSED METHODOLOGY

RoomMind AI will follow five main stages: dataset preparation, image preprocessing, CNN training, change detection, and web application development.

Figure 1. Proposed System Workflow

BEFORE and AFTER Images → Preprocessing and Alignment → Siamese CNN → Change Detection and Classification → Flask Backend → Web Interface

2.1 Dataset Preparation

The dataset will contain pairs of photographs showing the same room before and after a change. Images will be collected using a smartphone or camera. Additional synthetic examples may be created by modifying images to simulate changes.

Each image pair will be labelled as:

- **Added:** An object appears in the AFTER image.
- **Removed:** An object present in the BEFORE image is missing afterward.
- **Moved:** An object changes its position.
- **Unchanged:** No meaningful change has occurred.

Changed regions will be annotated using bounding boxes or masks where required. The dataset will be divided into training, validation, and testing sets using a proposed ratio of 70%, 15%, and 15%. Images from the same original scene will remain in the same subset to reduce data leakage.

2.2 Image Preprocessing and Alignment

OpenCV will be used to resize and normalise the photographs before they are passed to the model. ORB feature matching and RANSAC will be investigated to align corresponding scene features and reduce differences caused by camera movement.

Brightness adjustments and data augmentation, including small rotations and translations, may also be applied during training. These techniques are intended to help the model focus on meaningful changes rather than minor visual variations.

However, alignment may not work reliably when camera movement is substantial or when there are insufficient matching features.

2.3 Siamese CNN and Change Detection

A Siamese CNN will process the BEFORE and AFTER photographs through two matching branches that share the same weights. Each branch will extract visual features, which will then be compared to identify possible changes.

The network will generate a change map to indicate the locations of likely differences. A classification component will predict the relevant category using the prepared training labels.

The model will be trained from randomly initialised weights rather than pretrained CNN weights. Reliable classification of moved objects will require suitable annotations and, where necessary, matching the same object across both photographs.

The initial implementation will focus on change detection and classification. It will not guarantee the recognition of arbitrary object names or the detection of physical damage.

2.4 Training and Loss Functions

The model will be trained using labelled image pairs and their corresponding annotations. Binary cross-entropy may be used for changed-versus-unchanged pixel prediction, while categorical cross-entropy may be used for change classification. Dice loss may also be included to improve the detection of small changed regions.

The Adam optimizer will be considered for updating the model weights. Validation performance will be monitored during training, and the best-performing model checkpoint will be saved for use in the application.

The final architecture and training settings will depend on the dataset size and available computing resources.

2.5 Web Application Development

The frontend will be developed using HTML, CSS, and JavaScript. Users will be able to upload and preview the BEFORE and AFTER photographs before starting the analysis.

Flask will receive the images through a POST /analyze request, validate the uploaded files, preprocess the photographs, and run the trained model. The results will be returned in JSON format, including predicted categories and region coordinates or mask information.

The website will display the photographs side by side, highlight detected regions, and present a list of predicted changes. It will also display an appropriate message when no significant change is detected or when an uploaded file cannot be processed.

3. EVALUATION AND TESTING

The trained model will be evaluated using the held-out testing dataset. Several metrics will be used to measure its performance.

Metric	Purpose
Precision	Measures how many predicted changes are correct.
Recall	Measures how many actual changes are detected.
F1-score	Combines precision and recall.
Classification accuracy	Measures the percentage of correctly classified examples.
Intersection over Union (IoU)	Measures overlap between predicted and reference regions.
False-alarm rate	Measures how often unchanged scenes are incorrectly reported as changed.
Processing time	Measures the time required to analyse one image pair.

A basic OpenCV image-difference method will serve as the baseline. Both methods will be evaluated on the same testing examples where their outputs permit a fair comparison.

Testing will also verify that image uploads, preprocessing, model predictions, and result visualisation work correctly together.

Actual performance scores will be recorded after the experiments have been completed. No results will be assumed beforehand.

4. RESULTS AND DISCUSSION

The final results will depend on the completed model training and evaluation. The following table will be filled using measurements obtained from the testing process.

Metric	Siamese CNN	OpenCV Baseline
Precision	To be measured	To be measured
Recall	To be measured	To be measured
F1-score	To be measured	To be measured
Accuracy	To be measured, if applicable	To be measured, if applicable
IoU	To be measured, if applicable	To be measured, if applicable
False-alarm rate	To be measured	To be measured
Processing time	To be measured	To be measured

The comparison will help determine whether the trained CNN provides an improvement over basic image differencing. False detections caused by shadows and lighting variations, along with missed changes involving small objects, will also be examined.

The system has several limitations. Its performance may decrease when photographs contain unfamiliar objects, major camera movements, or significant lighting differences. Training a CNN from scratch may also require a sufficient number of labelled examples and suitable computing resources.

5. CONCLUSION

RoomMind AI aims to make comparing room photographs easier by combining image processing, a custom-trained Siamese CNN, and a web application. The system will attempt to identify added, removed, moved, and unchanged scenes or regions and highlight the results for the user.

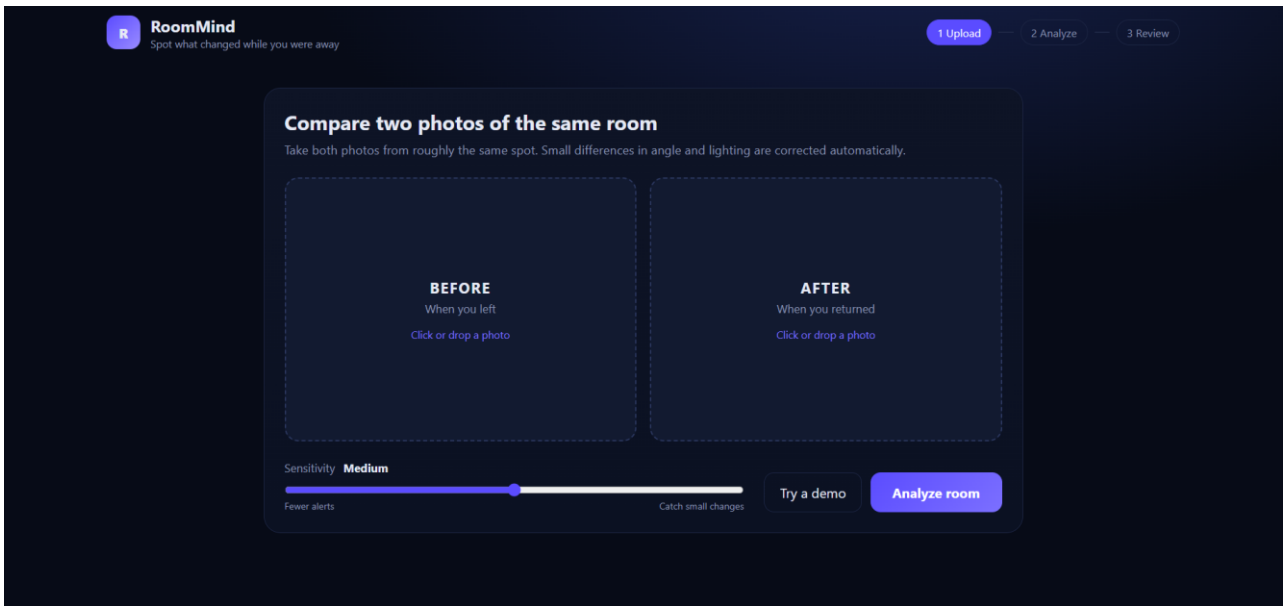
Image alignment and preprocessing will be used to reduce visual differences caused by camera movement and lighting. The trained model will then be evaluated against a basic OpenCV image-difference method to assess its usefulness.

The project's success will depend on the quality of the dataset, the training process, and the results obtained during testing. The completed application will provide a foundation for further improvements in object matching, change localisation, and robustness to different indoor conditions.

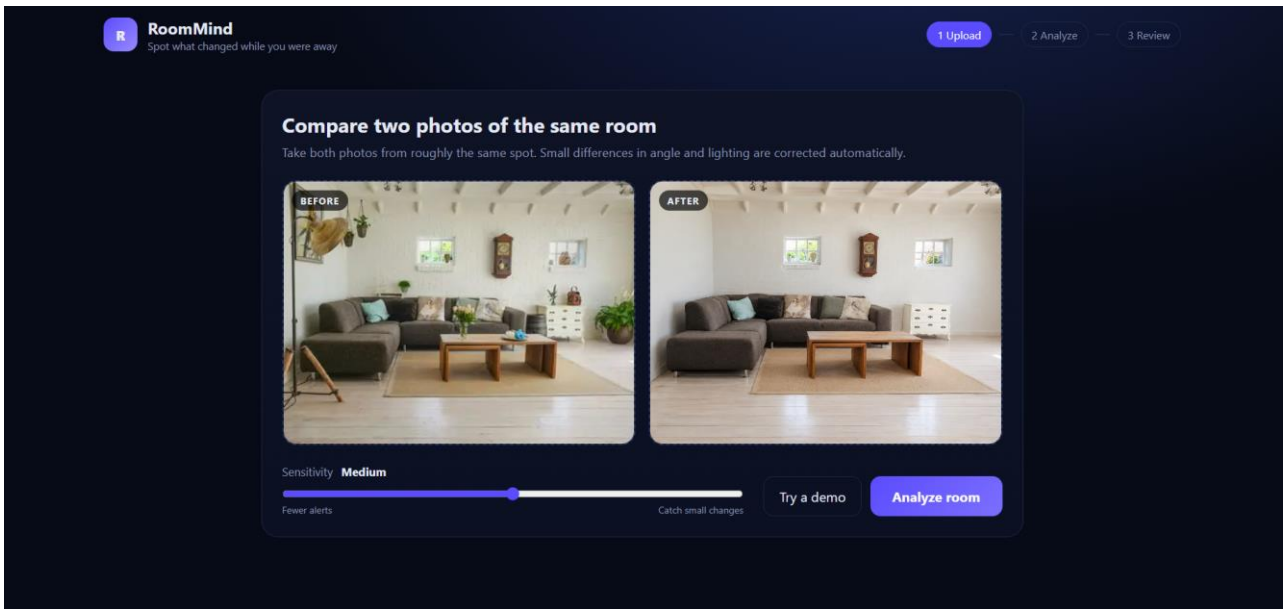
GitHub Repository Link: (Link)

6. Web Interface

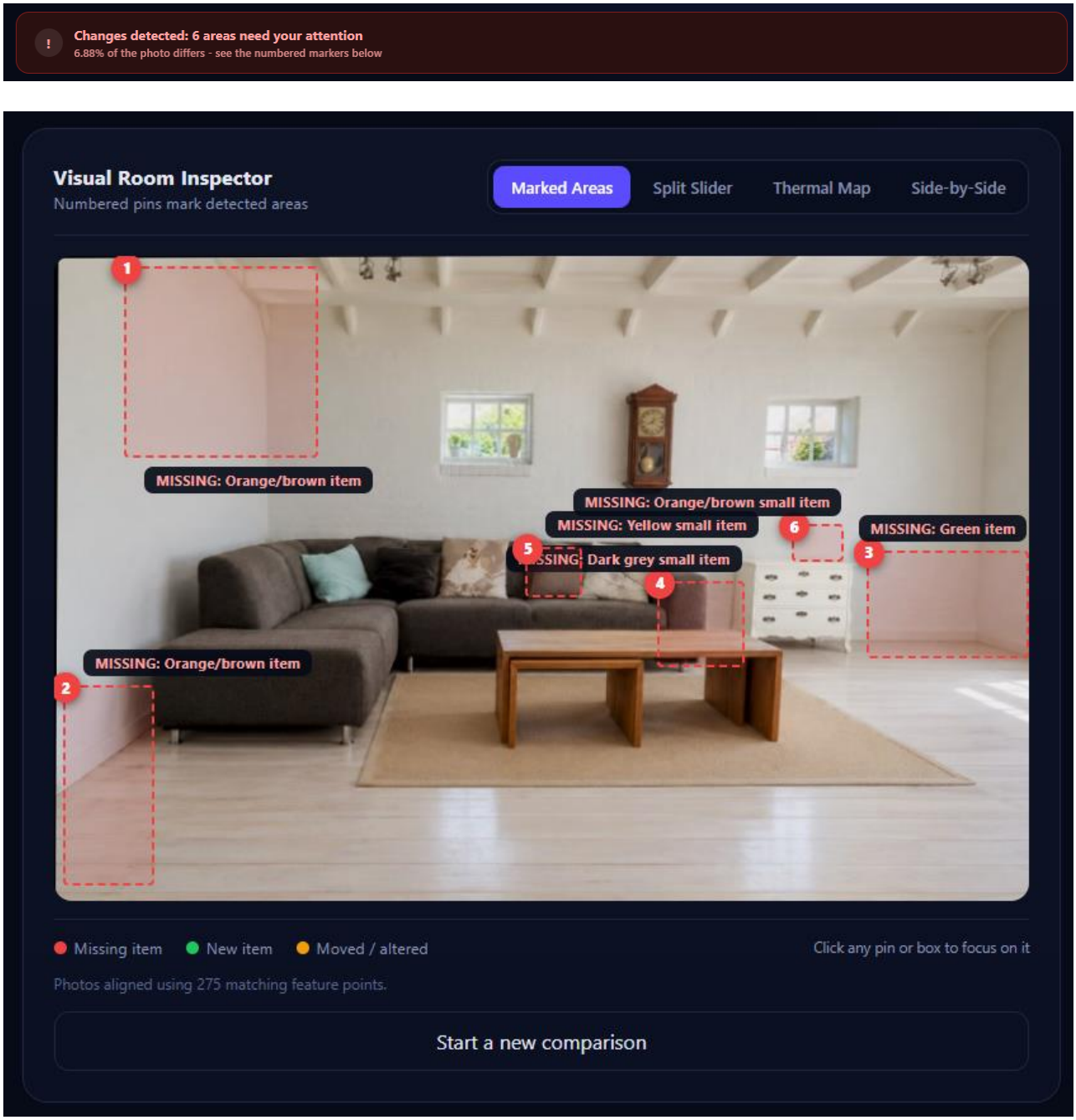
Interface:



Uploading Images:

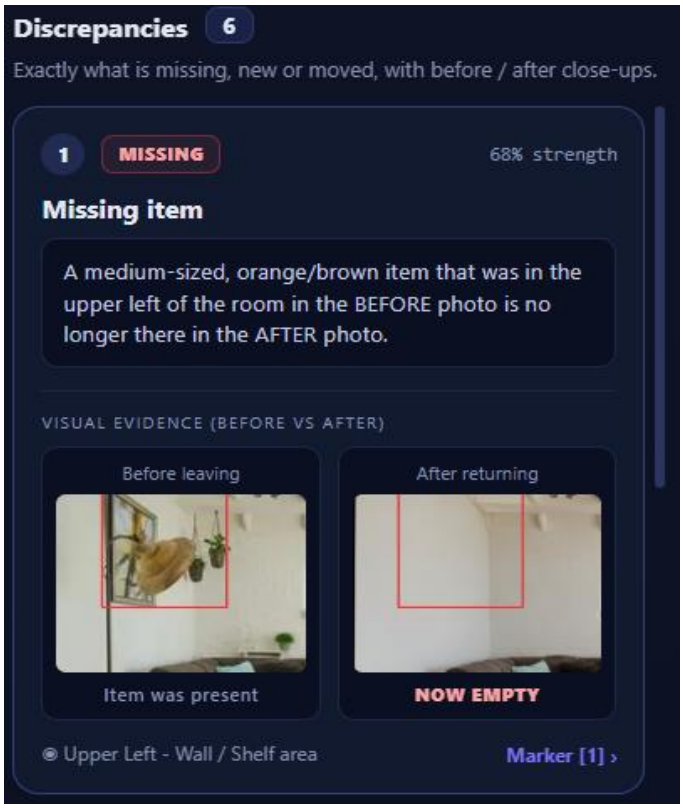


After Comparison:

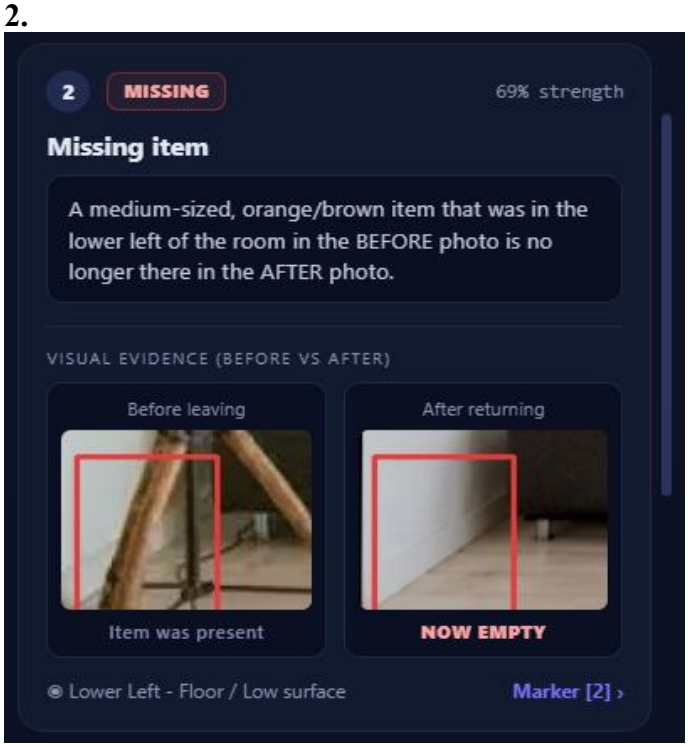


Detailed Reasons:

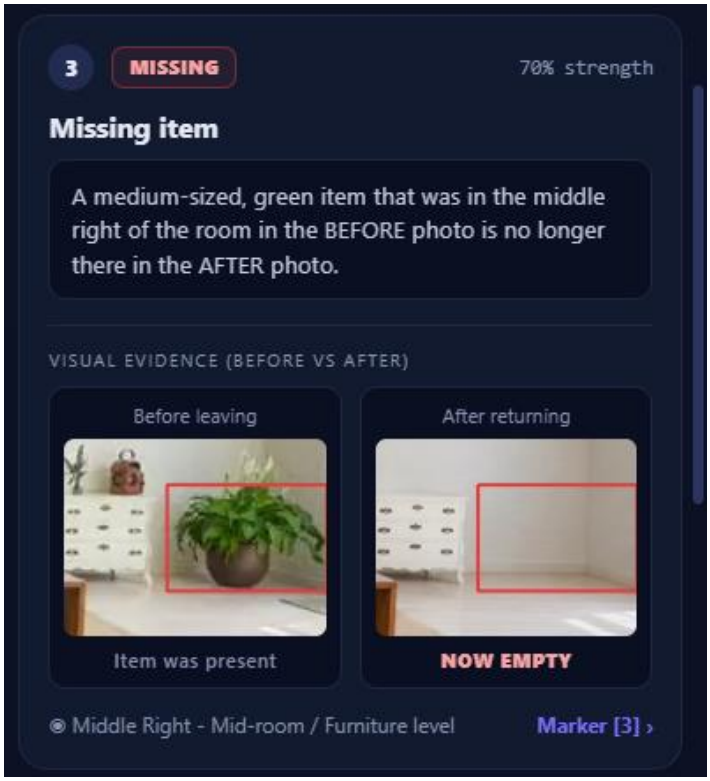
1.



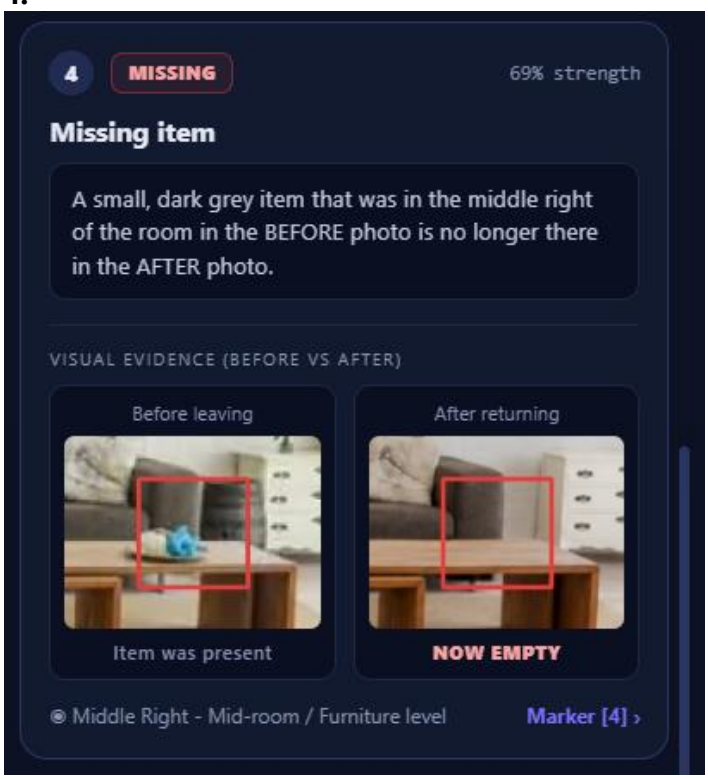
2.



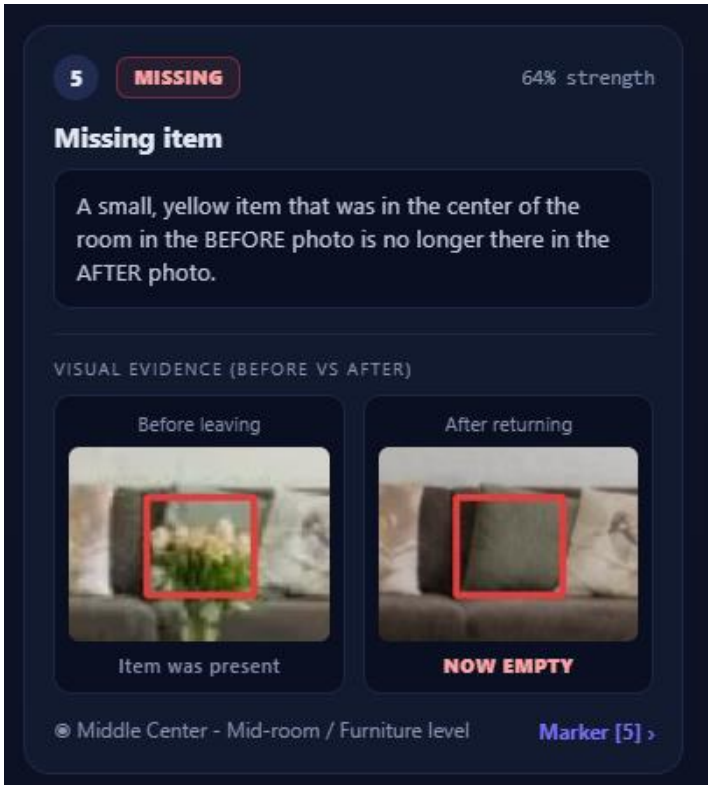
3.


3
MISSING
70% strength
Missing item
A medium-sized, green item that was in the middle right of the room in the BEFORE photo is no longer there in the AFTER photo.
VISUAL EVIDENCE (BEFORE VS AFTER)
Before leaving
After returning
Item was present
NOW EMPTY
Middle Right - Mid-room / Furniture level
Marker [3] ›

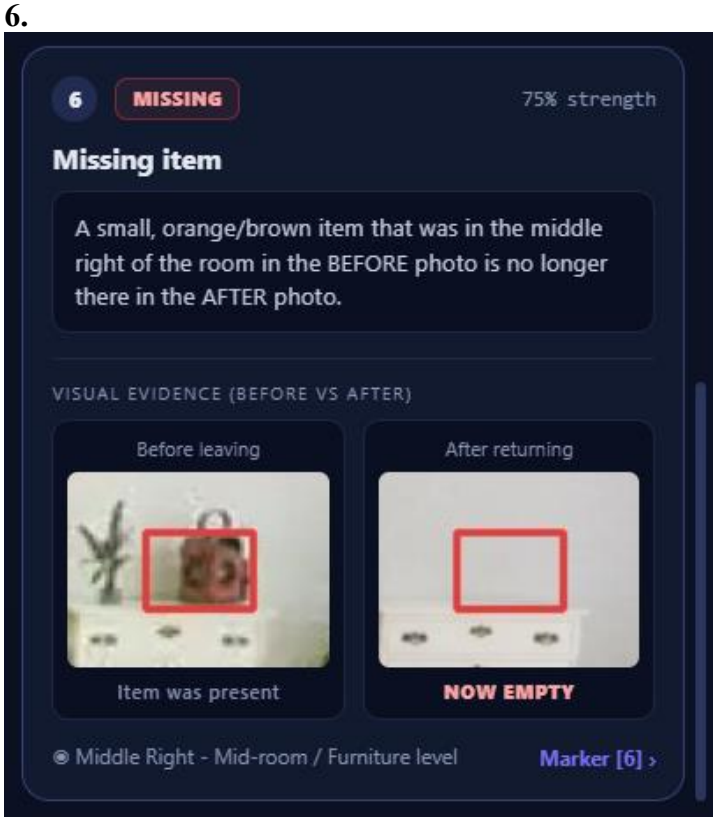
4.


4
MISSING
69% strength
Missing item
A small, dark grey item that was in the middle right of the room in the BEFORE photo is no longer there in the AFTER photo.
VISUAL EVIDENCE (BEFORE VS AFTER)
Before leaving
After returning
Item was present
NOW EMPTY
Middle Right - Mid-room / Furniture level
Marker [4] ›

5.


5
MISSING
64% strength
Missing item
A small, yellow item that was in the center of the room in the BEFORE photo is no longer there in the AFTER photo.
VISUAL EVIDENCE (BEFORE VS AFTER)
Before leaving
After returning
Item was present
NOW EMPTY
Middle Center - Mid-room / Furniture level
Marker [5] ›

6.


6
MISSING
75% strength
Missing item
A small, orange/brown item that was in the middle right of the room in the BEFORE photo is no longer there in the AFTER photo.
VISUAL EVIDENCE (BEFORE VS AFTER)
Before leaving
After returning
Item was present
NOW EMPTY
Middle Right - Mid-room / Furniture level
Marker [6] ›